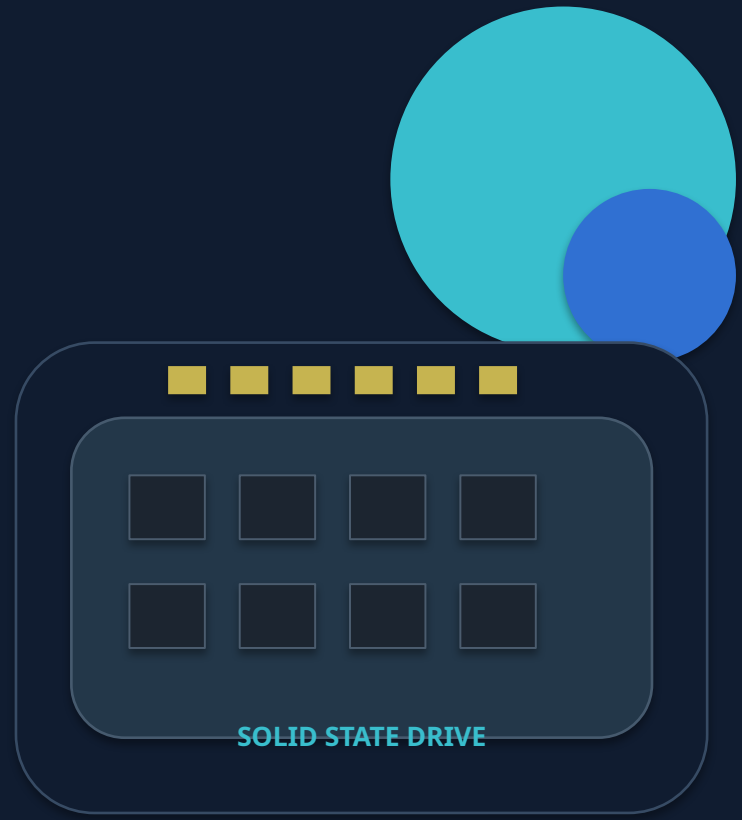
SSD

Solid State Drive

A modern secondary storage technology built for speed, efficiency and reliability.



1. What is an SSD?

A modern secondary storage device

- SSD stands for Solid State Drive.
- It is a type of secondary storage used to store data permanently.
- Commonly found in modern computers, laptops and other electronic devices.



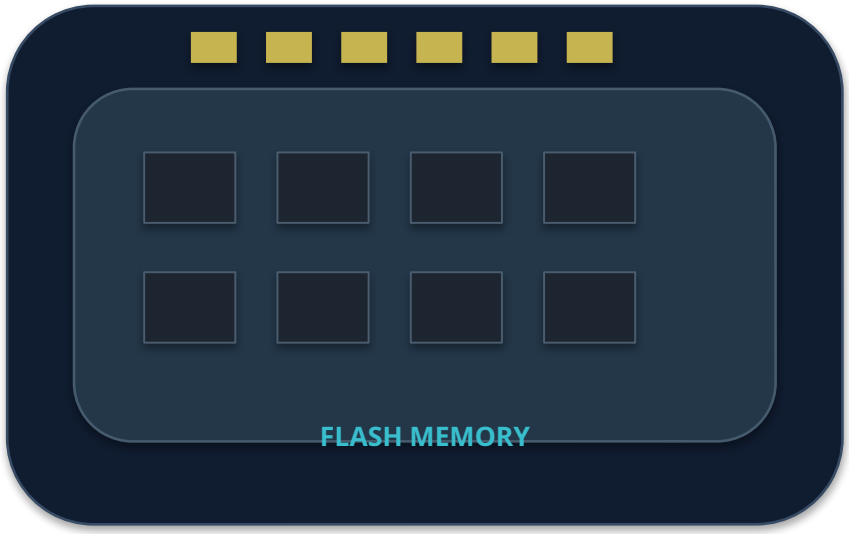
Permanent

Fast access

Flash memory

2. How SSDs Store Data

Flash memory replaces the mechanical parts found in HDDs



No spinning platters • No moving read/write head

- SSD → uses flash memory to store data.
- HDD → uses spinning platters and a moving read/write head.
- The absence of mechanical parts helps SSDs operate quietly and handle physical shocks better.

3. No Moving Parts

A solid-state design improves quiet operation and physical resistance



- No spinning disks.
- No moving read/write head.
- Less mechanical noise.
- Generally more resistant to physical shocks.

4. Speed

One of the biggest advantages of SSD technology



- Reads and writes data much faster than a traditional HDD.
- Helps a computer boot faster.
- Applications open more quickly.
- Files can be transferred more efficiently.

5. Advantages of SSD

FAST

Very fast performance

SILENT

Quiet operation

DURABLE

Better shock resistance

EFFICIENT

Usually lower power use

6. Disadvantages

The main limitation is cost



- SSDs are generally more expensive than HDDs for the same storage capacity.
- Large-capacity SSDs can therefore be costly.

7. Where SSDs Are Used

Fast storage is useful across many modern devices



Laptops — Fast everyday computing

Desktop PCs — Quick boot and app loading

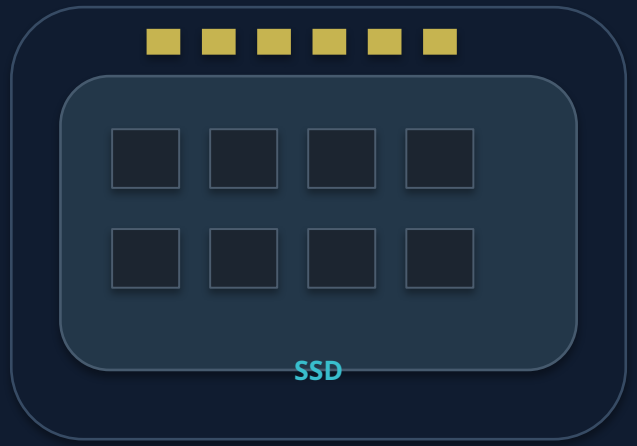
Gaming systems — Faster game and file loading

Other devices — High-speed storage needs

8. Conclusion

SSD is a modern and fast storage technology

- High speed and efficient data access
- Low power consumption
- Quiet operation
- Better resistance to physical shocks
- More expensive than HDD, but widely used in modern computers



THANK YOU